

Where deep learning losses come from I

Distributions, likelihood and loss: a model should say which values are possible and how plausible each one is.

THE QUESTION

Where does a training loss come from, before anyone writes one down?

THE INTUITION

Say what values the target can take and put a distribution over them; the network outputs its parameters. Fix the observed data and score each θ by the probability it gave them. Logs and a minus sign turn that score into a loss.

KEY TAKEAWAY

A loss is $-\log$ of the probability (or density) the model gave to the truth. Minimizing the mean NLL is maximum likelihood.

target & support → distribution family → network → parameters → likelihood of data → log → $-\log = \text{loss}$

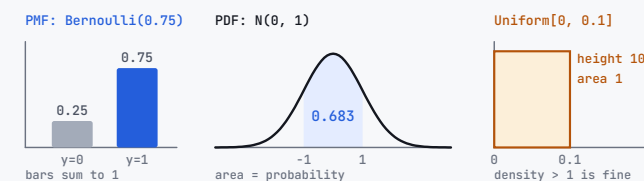
1 Start with the random variable

binary	● 0 ● 1	{0, 1}	Bernoulli
one of K	● 1 ● 2 ● 3 ... ● K	{1, ..., K}	Categorical
count	● 0 ● 1 ● 2 ● 3 ● 4 →	{0, 1, 2, ...}	Poisson
real	←————→	any real number	Gaussian

The support $\mathcal{S}_Y$ is the set of values the target can take. Match it to the task before choosing a loss.

Impossible targets should get zero probability or density. A support mismatch is a modelling error, not an optimizer problem.

2 Mass for discrete, density for continuous



PMF $p_Y(y) = P(Y = y)$ and $\sum_{y \in \mathcal{S}_Y} p_Y(y) = 1$

PDF $p_Y(y) \geq 0$ and $\int_{\mathcal{S}_Y} p_Y(y) dy = 1$

For continuous Y , $P(a \leq Y \leq b) = \int_a^b p_Y(y) dy$. A density value can exceed 1; only the area is a probability.

3 Four distribution families

family	parameters	support
Bernoulli	$p \in [0, 1]$	$\{0, 1\}$
Categorical	$p_1, \dots, p_K$	$\{1, \dots, K\}$
Poisson	$\lambda > 0$	nonnegative integers
Gaussian	$\mu \in \mathbb{R}, \sigma > 0$	$\mathbb{R}$

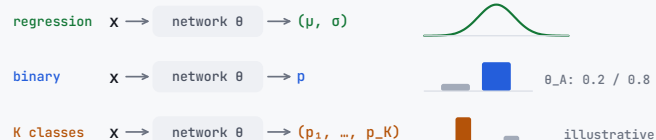
4 Read conditional model notation

$$p_\theta(y | x)$$

x one observed input
 y one possible or observed target
 θ all learned weights and biases
 $p_\theta(y|x)$ probability or density given to y , given x

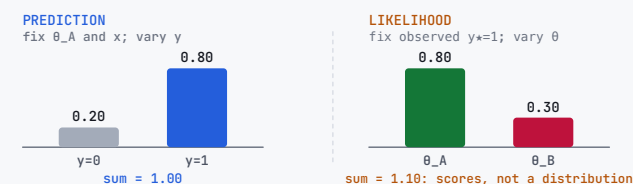
In supervised training the observed inputs are treated as fixed; the model describes the conditional target distribution.

5 The network predicts distribution parameters



$x \rightarrow (\mu_\theta(x), \sigma_\theta(x))$ for regression, $x \rightarrow p_\theta(x)$ for a binary label, and $x \rightarrow (p_{\theta,1}(x), \dots, p_{\theta,K}(x))$ for K classes. A point prediction is only a summary (often the conditional mean or mode). Training scores the full distribution at the observed target.

6 Prediction and likelihood ask different questions



Prediction: fix θ and x ; vary the possible outcome y .
Likelihood: fix the observed data; vary candidate parameters θ .

The same expression $p_\theta(y|x)$ is read differently because a different quantity varies.

7 Pause and think: what does each choice commit you to?

Why pick the support first? A loss can only score outcomes the distribution allows. A Gaussian head for a count puts density on -3.5 .

Can a density equal 10? Yes: Uniform[0, 0.1] has height 10. Only the area over an interval must be at most 1.

Does likelihood sum to 1 over θ ? No: $0.80 + 0.30 = 1.10$. It is a function of θ , not a distribution over θ .

Choose the target, then its support, then a distribution family, then the neural outputs that parameterize it.

Build a dataset likelihood

Likelihood ranks parameter settings by how well they explain the outcomes that actually occurred.

8 Score one fixed observation

For one pair (x^*, y^*) the likelihood contribution is $L_i(\theta) = p_\theta(y^* | x^*)$. A larger value means that candidate θ explains this fixed observation better.

For a continuous target this is a density. Comparing candidates is still valid, even though a density is not itself an event probability.

9 Independence gives a product

flips H H T T T H H T T T

$\theta = 0.4$ $0.4 \times 0.4 \times 0.6 \times 0.6 \times 0.6 \times 0.4 \times 0.4 \times 0.6 \times 0.6 \times 0.6$
 $\theta = 0.7$ $0.7 \times 0.7 \times 0.3 \times 0.3 \times 0.3 \times 0.7 \times 0.7 \times 0.3 \times 0.3 \times 0.3$

$L(0.4) = 0.00119$ $6.8\times$ larger $L(0.7) = 0.000175$

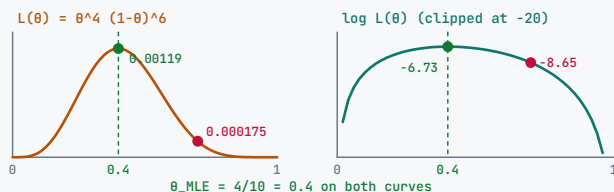
For conditionally independent examples, $L(\theta) = p(\mathcal{D}|\theta) = \prod_{i=1}^n p_\theta(y_i | x_i)$.

Independent: multiply the per-example factors.

Identically distributed: reuse the same conditional model form.

These are separate assumptions; either can fail without the other. Above: ten flips with $P(H) = \theta$ give $L(\theta) = \theta^4(1-\theta)^6$.

10 MLE names the winner, and logs keep it



$\hat{\theta}_{MLE} = \arg \max_{\theta} L(\theta)$: the parameters that give the observed data the largest likelihood. Since log is strictly increasing on positive values, $\arg \max_{\theta} L = \arg \max_{\theta} \log L$, and a product becomes a sum: $\log L(\theta) = \sum_i \log p_\theta(y_i | x_i)$. Easier derivatives, and no underflow: 200 factors of 0.01 multiply to exactly 0.0 in float64 (from factor 162 on), while the log-sum is -921.03.

11 In code: the loss is a mean of $-\log_prob$

```
flips = torch.tensor([1,1,0,0,0,1,1,0,0,0])
for th in (0.4, 0.7): # H = 1
    d = Bernoulli(probs=th)
    nll = -d.log_prob(flips).mean()
# 0.673 at 0.4 < 0.865 at 0.7
```

.sum() gives $6.73 < 8.65$ instead: the same winner, on a scale n times larger.

12 Tiny Bernoulli comparison

Compare $\theta = 0.5$ and $\theta = 0.8$, where $P(H|\theta) = \theta$:

data	$L(0.5)$	$L(0.8)$
H, T	$0.5 \cdot 0.5 = 0.25$	$0.8 \cdot 0.2 = 0.16$
H, H	$0.5^2 = 0.25$	$0.8^2 = 0.64$

The ranking of candidates changes when the fixed observed data change.

13 Negative log-likelihood is a loss

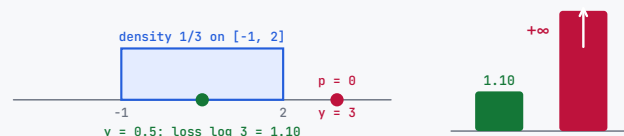


Optimizers minimize, so negate: $\ell_i(\theta) = -\log p_\theta(y_i | x_i)$ and

$$\hat{R}(\theta) = \frac{1}{n} \sum_i \ell_i(\theta) = -\frac{1}{n} \log L(\theta).$$

The factor $1/n$ changes the gradient scale, not the minimizer.

14 Support becomes a hard constraint



If the model gives an observed target $p_\theta(y_i | x_i) = 0$, then $\ell_i = -\log 0 = +\infty$. A zero-likelihood observation is an infinite penalty. Check the support before blaming numerical instability.

15 Final audit

- ✓ Target support matches the chosen distribution.
- ✓ Neural outputs obey the parameter constraints.
- ✓ Each observation contributes one log-density or log-probability.
- ✓ Independence assumptions are stated, not silently assumed.
- ✓ Likelihood and prediction are not confused.

16 Pause and think

Why is $\hat{\theta}_{MLE} = 0.4$? $d/d\theta \log L = 4/\theta - 6/(1-\theta) = 0$ gives $4/10$: the observed fraction of heads.

Mean or sum of NLL? Dividing by $n > 0$ rescales the loss but never reorders candidates.

Probability models possible outcomes · likelihood scores θ by what occurred · $-\log$ turns it into a loss

Numbers from the Lecture 1 storyboard data (the deck's ten coin flips H H T T T H H T T T; displayed values rounded). Slides, notebooks and the summary video: nipunbatra.github.io/dl-2026.